

Sharif Uddin

Electrical Engineering | RF, Microwave & Antenna Research

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[Google Scholar](#) | [Website](#) | [LinkedIn](#)

Research Interests

RF and Microwave Engineering; Antenna Design and Characterization; Circularly Polarized Antennas; Full-Duplex Wireless Systems; MIMO Antennas and Mutual Coupling Reduction; mmWave and Terahertz Antennas; Wideband Antennas; Antenna Fabrication and Measurement; Electromagnetic Simulation and Computational Design.

Education

M.S. in Electrical Engineering

Washington State University Vancouver, USA

Fall 2025–Present

GPA: [X.XX/4.00]

- Research focus: Wideband circularly polarized antennas for full-duplex wireless communication.
- Advisor: Dr. Tutku Karacolak.

M.Sc. in Electrical, Electronic and Communication Engineering–Graduate Coursework Completed

Military Institute of Science and Technology (MIST), Bangladesh

[2022]–[2024]

CGPA: 3.58/4.00

- Coursework completed; degree not completed

B.Sc. in Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology (RUET), Bangladesh

[2016]–[2021]

CGPA: 3.54/4.00

Research Experience

Graduate Researcher

Washington State University Vancouver, USA

Fall 2025–Present

- Design and investigate **differentially fed wideband circularly polarized antennas** for full-duplex & wireless relay communication systems.
- Develop compact multi-port antenna architectures with emphasis on **port isolation, mutual coupling reduction, impedance matching, axial ratio, gain, and radiation efficiency**.
- Perform full-wave electromagnetic simulations using **ANSYS HFSS 2025 R1**, followed by antenna fabrication and experimental validation.
- Characterize fabricated antennas using microwave measurement equipment, including **Keysight FieldFox N9952A** and vector network analyzer measurements.

Publications

Published

- [1] **S. Uddin**, et al., “A Compact Wideband Circularly Polarized MIMO Antenna for High Data Rate Communications,” 2026 *National Radio Science Meeting (NRSM)*, 2026.
- [2] **S. Uddin**, et al., “An Improved Switching Control Technique of a Three-Phase NPC Multilevel Converter in Reducing the Neutral Point Potential Swing,” *IEEE Xplore*, 2024.
- [3] A. Hossain **S. Uddin**, et al., “Wavelet and Spectral Analysis of Normal and Abnormal Heart Sound for Diagnosing Cardiac Disorders,” *BioMed Research International*, 2022.

Manuscripts Under Review

- [1] **S. Uddin**, et al., “Differentially Fed Wideband Circularly Polarized Antenna for Full-Duplex Communications,” manuscript submitted for publication, 2026.

Selected Research Projects

Differentially Fed Wideband Circularly Polarized Antenna for Full-Duplex Communication

- Investigating wideband CP antenna architectures for simultaneous transmit and receive operation.
- Studying isolation mechanisms including polarization, spatial/radiation-pattern, structural, and decoupling-based techniques.
- Experimental validation through fabricated prototypes and measured S-parameters, radiation characteristics, gain, and efficiency.

Antenna Design, Simulation, Fabrication and Measurement

- Designed and optimized wideband and circularly polarized antennas using full-wave electromagnetic simulation.
- Fabricated antenna prototypes and performed experimental characterization using microwave/RF instrumentation.

Teaching Experience

Graduate Teaching Assistant, Department of Electrical & Computer Engineering <i>Washington State University Vancouver, USA</i>	[2025]–[Present]
Lecturer, Department of Electrical & Electronic Engineering <i>Northern University Bangladesh, Bangladesh</i>	[2023]–[2025]
• Taught undergraduate courses in electrical and electronic engineering and supervised undergraduate design and research projects.	
Lecturer, Department of Electrical & Electronic Engineering <i>Prime University, Bangladesh</i>	[2021]–[2023]

Technical Skills

Electromagnetics	Antenna design, RF/microwave engineering, CP antennas, MIMO, mutual coupling, full-duplex systems, mmWave/THz concepts
Simulation	ANSYS HFSS 2025 R1, CST Studio Suite
Measurement	VNA, Keysight FieldFox N9952A, S-parameter measurement, radiation pattern, gain and efficiency characterization
Fabrication	LPKF PCB/laser milling, antenna prototyping, laboratory fabrication and measurement
Data Analysis	MATLAB, Machine Learning, Design-Expert v13, DOE, ANOVA and experimental optimization
Programming	MATLAB, Python, C

Professional Activities

- Presented research at the 6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), 2024.
- Presented research at the National Radio Science Meeting (NRSM), 2026.

Additional Information

English Proficiency: IELTS Overall 7.0 (L 8.0, R 7.0, W 7.0, S 6.5)

GRE: Quantitative 159, Verbal 144, AWA 3.5

References

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